HAONAN CHEN

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EDUCATION

Ph.D. in Electrical and Computer Engineering
University of Illinois Urbana-Champaign 08/2020 - Present

B.Sc. in Electrical Engineering with Highest Honors
University of Illinois Urbana-Champaign 09/2016 - 05/2020
GPA: 3.98/4.0

B.Eng. in Electrical Engineering and Automation
Zhejiang University 09/2016 - 06/2020
GPA: 3.99/4.0 (Rank: 1/30)

UNDER REVIEW

H. Chen, C. Zhu, Y. Li, K. Driggs-Campbell. "Tool-as-Interface: Learning Robot Tool Use from Human Play through Imitation Learning." *Under Review,* 2025. [pdf] [Website]
Best Paper Award at [ICRA 2025 workshop on Foundation Models and Neural-Symbolic (NeSy) AI for Robotics]
Best Presentation Award at [CSL Student Conference 2025]

JOURNAL PUBLICATIONS

[ITSM'24] A. Hasan, N. Chakraborty, H. Chen, J.-H. Cho, C. Wu, K. Driggs-Campbell. "Lessons in Cooperation: A Qualitative Analysis of Driver Sentiments towards real-time Advisory Systems through a Focus Group User Study." *IEEE Intelligent Transportation Systems Magazine,* 2024. [pdf]

[JATS'24] A. Hasan, N. Chakraborty, H. Chen, J.-H. Cho, C. Wu, K. Driggs-Campbell. "Cooperative Advisory Residual Policies for Congestion Mitigation." *ACM Journal on Autonomous Transportation Systems*, 2024. [pdf]

PEER REVIEWED CONFERENCE PUBLICATIONS

[RSS'25] S. Peng, H. Chen, K. Driggs-Campbell. "Towards Uncertainty Unification: A Case Study for Preference Learning." *Robotics: Science and Systems,* 2025. [pdf] [Website]

[ICRA'25] H. Chen, J. Xu*, L. Sheng*, T. Ji, S. Liu, Y. Li, K. Driggs-Campbell. "Learning Coordinated Bimanual Manipulation Policies using State Diffusion and Inverse Dynamics Models." *IEEE International Conference on Robotics and Automation,* 2025. [pdf] [Website]

[CoRL'23] H. Chen, Y. Niu*, K. Hong*, S. Liu, Y. Wang, Y. Li, K. Driggs-Campbell. "Predicting Object Interactions with Behavior Primitives: An Application in Stowing Tasks." *Conference on Robot Learning.* [pdf] [Website]
Finalist - Best Paper/Best Student Paper Awards

[CASE'23] Z. Huang, Y. Mun, H. Chen, Y. Xie, Y. Niu, X. Li, N. Zhono, H. You, D.L. McPherson, K. Driggs-Campbell. "Towards Safety of Multi-Level Human-Robot Interaction in Industrial Tasks." *IEEE Conference on Automation Science and Engineering (CASE), Special Session on Cyberphysical Manufacturing Networks,* 2023. [pdf]

[**CASE'23**] H. Chen*, Y. Mun*, Z. Huang, Y. Niu, Y. Xie, D.L. McPherson, K. Driggs-Campbell. "Learning Task Skills and Goals Simultaneously from Physical Interaction." *IEEE Conference on Automation Science and Engineering (CASE), Special Session on Cyberphysical Manufacturing Networks,* 2023. [pdf]

[**ITSC'23**] A. Hasan, N. Chakraborty*, H. Chen*, J-H. Cho, C. Wu, K. Driggs-Campbell. "PeRP: Personalized Residual Policies For Congestion Mitigation Through Co-operative Advisory Systems." *IEEE International Conference on Intelligent Transportation Systems,* 2023. [pdf] [Website]
Best Presentation Award at [CSL Student Conference 2024]

[**ITSC'22**] H. Chen, T. Ji, S. Liu, K. Driggs-Champbell. "Combining Model-Based Controllers and Generative Adversarial Imitation Learning for Traffic Simulation." *IEEE International Conference on Intelligent Transportation Systems (ITSC 2022).* [pdf]

[**ICRA'22**] S. Liu, P. Chang, H. Chen, N. Chakraborty, K. Driggs-Champbell. "Learning to Navigate Intersections with Unsupervised Driver Trait Inference." *IEEE International Conference on Robotics and Automation,* 2022. [pdf] [Website]

[**IROS'20**] P. Chang, S. Liu, H. Chen, K. Driggs-Campbell. "Robot Sound Interpretation: Combining Sight and Sound in Learning-Based Control." *IEEE/RSJ International Conference on Intelligent Robots and Systems,* 2020. [pdf] [Video]

INVITED TALKS & PANELS

April 2025 Professor Anima Anandkumar 's Lab, Caltech
Title: *Deployable Robot Learning from Human Priors*

April 2025 Professor Katerina Fragkiadaki's Lab, Carnegie Mellon University
Title: *Deployable Robot Learning from Human Priors*

March 2025 Safe AI Lab, Carnegie Mellon University
Title: *Building Deployable and Pragmatic Robots from Human Priors*

March 2025 Parasol Lab, University of Illinois Urbana-Champaign
Title: *Tool-as-Interface: Learning Robot Tool Use from Human Play through Imitation Learning*

TEACHING EXPERIENCE

Graduate Teaching Assistant

- **UIUC ECE 484:** Safe Autonomy — **Spring 2025**
  Instructor: Prof. Sayan Mitra

Undergraduate Teaching Assistant

- **ZJU-UIUC Math 241:** Calculus III — **Fall 2018**
  Instructor: Prof. Thomas Honold
- **ZJU-UIUC ECE 310:** Digital Signal Processing
  ZJU-UIUC ECE 311: Digital Signal Processing Lab — **Fall 2019**
  Instructor: Prof. Mark D. Butala

MENTORED STUDENTS

M.S. Students

- Yiyang Xu

Undergraduate Students

- Jiaming Xu — Placement: University of Illinois Urbana-Champaign

- Lily Sheng — Placement: Tsinghua University
- Yilong Niu — Placement: University of Illinois Urbana-Champaign
- Cheng Zhu — Placement: University of Pennsylvania
- Hanlin "Asher" Mai — Placement: University of Illinois Urbana-Champaign
- Simon Xia
- Hyoungju Lim

AWARDS AND HONORS

Conference Travel Award, Graduate College, University of Illinois Urbana-Champaign 2023
Finalist - Best Paper/Best Student Paper Awards, Conference on Robot Learning 2023
Bronze Tablet Award, University of Illinois Urbana-Champaign 2020
Excellent Graduate, Zhejiang University 2020
Student Speaker at Graduation, ZJU-UIUC Institute 2020
Provincial Government Scholarship, Zhejiang Province, China (Top 3%) *2019*
Dean's List, University of Illinois Urbana-Champaign 2019
National Scholarship, China (Top 1.8%) 2018
Excellent Undergraduate Scholarship, Zhejiang University (Top 3%) 2018-2019
Academic Excellence Scholarship Award, ZJU-UIUC Institute 2017-2019
Top Ten Social Practice, Zhejiang University (Top 1.5%) 2017

SERVICE AND LEADERSHIP

- **ICRA 2025 Workshop on Learning Meets Model-Based Methods for Contact-Rich Manipulation** — Lead Organizer · May 2025
  Workshop Website
  Successfully Raised $1,400 in sponsorship from three companies and three IEEE RAS Technical Committees. The workshop attracted over 300 attendees.
- **Robotics Seminar @ Illinois** — Student Co-Organizer · August 2023–Present
  Seminar Series
- **Coordinated Science Laboratory Student Conference** — Reception Chair · 2024
  Conference Website

PROFESSIONAL SERVICE

- **Session Chair:**
  - IEEE International Conference on Robotics and Automation (ICRA), 2025 – Language-Guided Manipulation
  - CASE - IEEE Conference on Automation Science and Engineering, 2023
- **Reviewer:**
  - CoRL - Conference on Robot Learning
  - ICRA - IEEE International Conference on Robotics and Automation
  - ITSC - IEEE International Conference on Intelligent Transportation Systems
  - CASE - IEEE Conference on Automation Science and Engineering

OUTREACH AND EXHIBITIONS

Sweet Gestures: Interactive Cupcake Sprinkling Robot 2024

- **Engineering Open House 2024** (Urbana-Champaign, IL)
  Designed and showcased robotic exhibit for 20,000+ attendees, engaging K-12 students from across Illinois
- **Society of Women Engineers WE24 Conference** (Chicago, IL)
  Demonstrated interactive robotics platform to 11,000+ conference participants

STEM Talk: Toward Human-Robot Alignment through Human Behavior Priors May 2025
Presented to Grade 5 students at St. Matthew Catholic School, fostering early interest in robotics and AI through accessible storytelling.

EXTRA-CURRICULAR SERVICE

Founding Team Leader, Robotic Team for RoboMaster Challenge, ZJU-UIUC Institute 10/2017-06/2018

REFERENCES

Katie Driggs-Campbell
Assistant Professor, University of Illinois Urbana-Champaign
krdc@illinois.edu
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Yunzhu Li
Assistant Professor, Columbia University
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